SCIENTIFIC SECTION

MAIN LANGUAGE SELECTED = ENGLISH

Type of instrument	Research project (PDR)
Full name of the main promoter	ERNST Damien
Shortened title of the project (40 characters max., including spaces)	GBOML for uncertainty and bilevel opt.
Reference e-space	40021584

Remarks:

1) *The main promoter is the responsible proponent accountable to the F.R.S.-FNRS administration and must fill in the sections here below and convert the file into an unprotected PDF before appending it to the electronic form.*

2) *The F.R.S.-FNRS insists on* ***strict compliance with the instructions given for each part of the proposal*** *(scientific section relevant to the instrument selected, number of pages allowed for documents to be enclosed with the application form…) and stresses again the sovereign consideration of the Scientific Commissions assessing the application file.*

3) *In case of mismatch between the 'scientific section' and the online form, only the information provided in the form will be considered for the review of the research project.*

1) BRIEF REPORT ON PREVIOUS WORKS

Please write a brief report (max. 2 pages) on your previous works, drawing connections with your new project and mentioning your significant references on the subject (please make a list in the form of footnotes).

1.1 The Graph-Based Optimization Modeling Language (GBOML)

Over the past three years, our team has been involved in the INTEGRATION project, which is funded by the SPF Economie. One of our objectives has been to develop a methodology for modelling and optimizing energy system sizing and operations problems within the context of the ongoing Belgian energy transition. Energy system sizing/investments and operational problems can be viewed as networks of interconnected technologies or units, as shown in Figure 1, for which it is important to determine the best investments to be made (e.g., size of the PV installation and of the batteries) and the best operational strategy to implement. They can be effectively addressed through Mixed-Integer Linear Programming (MILP). We formulate these problems using variables, constraints, and objectives.

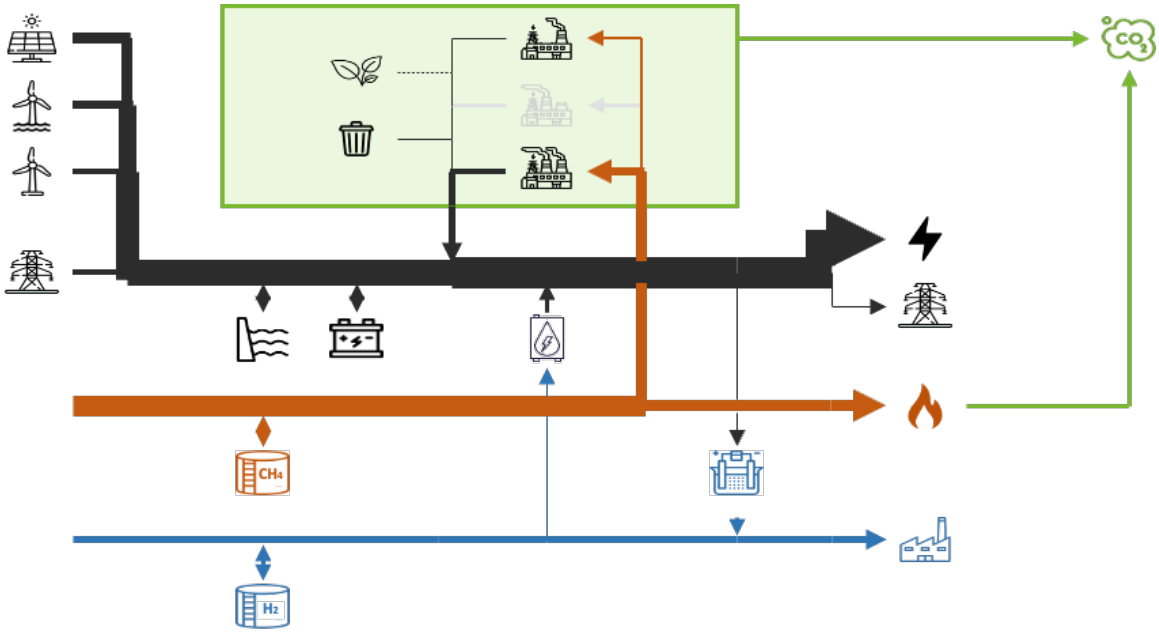

Figure 1: The Belgian energy system as seen in the context of INTEGRATION. The goal is to design and optimize a modelling system in order to minimize the cost (both operational and investment) in order to achieve net-zero carbon emissions in 2050.

We have created the Graph-Based Optimization Modeling Language (GBOML) in order to capture and exploit the structure that exists in these problems. GBOML is a modelling language made for structured MILPs. This modelling language leverages the inherent network structure of the problems from model encoding to output generation. GBOML allows users to encode MILPs using a hierarchical hypergraph (shown in Figure 2), ensuring that all essential features of the model are captured in its fully symbolic internal structure. By utilizing the existing structure, GBOML enables parallel problem instance generation, facilitates model and component reuse, and supports component assembling.

GBOML interfaces with Dantzig-Wolfe and Benders algorithms, two structure-exploiting methods. Furthermore, it produces structured outputs. GBOML has been implemented in Python, is thoroughly documented, and can be easily installed via PyPI, the Python online library. It is released under the MIT open-source license allowing anyone to download/use/modify/enhance the tool.

In terms of industry impact, GBOML has gained significant traction, with over 8,000 downloads[1]. It is currently being used by two companies, Fluxys and Haulogy, and has attracted the interest

[1] as reported on https://pepy.tech/project/gboml

of other major companies including Anglo-American, TotalEnergies, and TES-H2. Moreover, it is actively employed in numerous projects associated with the Smart Grids Lab at ULiège. Our contributions have also been recognized in the scientific community, with one published paper[2], another under review[3] and four oral presentations at international scientific conferences[4] and workshops.

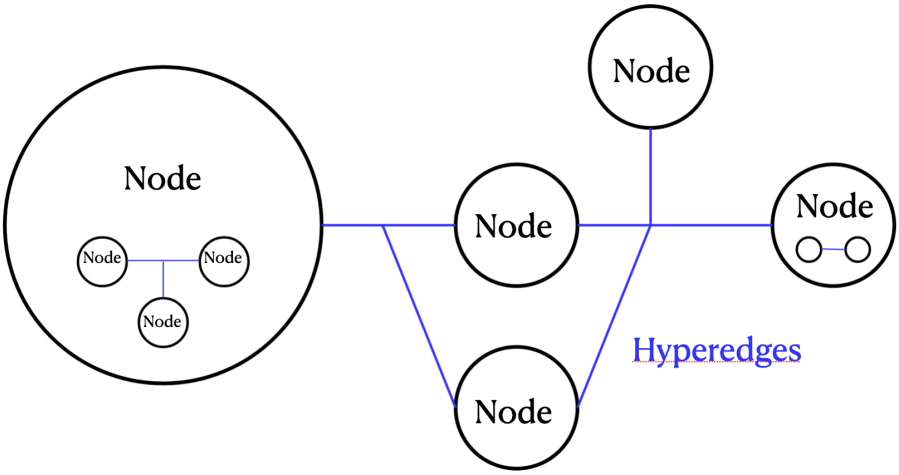

Figure 2: The hierarchical hypergraph structure.

While GBOML has already proven its value, we believe that its structured symbolic framework can be further exploited. There are several potential avenues for enhancement and exploration. Firstly, one of the main issues with MILP is how to take into account the uncertainty that might exist over certain parameters being considered, and consequently we look for solutions which will take into account that uncertainty. Afterwards, we want to explore how our framework can enable bi-level optimization. Finally, we will aim to investigate the capability of the existing structure to automatically identify optimal decompositions for the Dantzig-Wolfe and Benders algorithms, thus further improving their efficiency and effectiveness.

1.2 Other related works

Guillaume Derval, one of the project contributors, has key experience in modeling languages, having worked on OscaR[5], a constraint programming (CP) language used widely in the CP community. He also received the ORBEL-Wolsey award (highlighting the best open-source contribution in the field of Operations Research in Belgium) for OscaR-Modeling[6], a symbolic first modelling language above OscaR.

A first attempt to handle uncertainty within a context similar to one being handled by GBOML (combined investment and operations problems) was made using the Reinforcement Learning paradigm in the paper “Jointly Learning Environments and Control Policies with Projected Stochastic Gradient Ascent” (Bolland, A., Boukas, I., Berger, M., & Ernst, D., Artificial Intelligence Research, 2022).

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[2] Miftari, B., Berger, M., Djelassi, H., & Ernst, D. . GBOML: Graph-Based Optimization Modeling Language. Journal of Open Source Software, 7(72), 4158, https://doi.org/10.21105/joss.04158

[3] Miftari, B., Berger, M., Derval, G., Louveaux, Q., & Ernst, D. (2022). GBOML: A Structure-exploiting Optimization Modelling Language in Python. Under review at the Optimization Methods and Software journal.

[4] B. Miftari, G. Derval, and D. Ernst. “Tutorial: “An Introduction To Sizing And Operations of Energy Systems with GBOML”.” Oral presentation at 2nd International workshop on “Open Source Modeling and Simulation of Energy Systems” OSMSES 2023, Aachen, Germany, 2023.

B. Miftari, M. Berger, G. Derval, and D. Ernst. “Exploiting structure in MILP: a modeler’s perspective.” Oral presentation at SIAM OP23, Seattle, United States, 2023.

Miftari, B., Berger, M., Derval, G., Louveaux, Q., & Ernst, D. (2023). GBOML: A Structure-exploiting Optimization Modelling Language in Python. Poster presented at MIP2023, Los Angeles, United States, 2023.

Miftari, B., Berger, M., Derval, G. & Ernst, D. (2023). GBOML: A modelling tool for structured MILPs. Accepted oral presentation at IFORS2023, Santiago, Chile, 2023.

[5] https://bitbucket.org/oscarlib/oscar/src/dev/ - Google Scholar Link

[6] https://www.orbel.be/activities-wolsey.php - Related paper: Embarassingly Parallel Search Reengineered (CP2023 – Doctoral Program)

2) DESCRIPTION OF THE RESEARCH PROJECT

The written project must be made up of 4 parts (max. 4 pages) according to the structure below, accompanied by a reference bibliography (max. 1 page besides the 4 pages dedicated to the project) listed by order of appearance within the text.

Graphs and tables may be added (max. 2 pages).

2.1 Goals of the research

As explained in section 1, GBOML allows for the modelling of MILPs and focuses on reusability; this was achieved through the use of symbolism (that is, constraints and variables can be reworked before being submitted to an underlying solver). This symbolism opens up new possibilities, on which this research project focuses. GBOML does not live in isolation; it is deployed by real-life users (such as our lab, Fluxys and Haulogy), which have uses related to energy transition. **One ambition of this research project is for GBOML to become one of the go-to tools when modelling large energy systems.** For this, we propose exploring in three interlinked directions, each allowing for the modelling of more complex problems, solving them faster, or gaining additional insights into the results of the modelling:

- Managing uncertainty, using multi-parametric, stochastic and robust optimization
- Bi-level optimization
- Optimal structure finding to make the underlying solver faster

Not only do we want to integrate these capabilities in GBOML, but also **create new methods suited to the needs of our problems and users**. We describe precisely what this entails in section 2.3 (Research Project).

As an aside to this research, the PDR will also contribute to maintaining GBOML as a high-quality open-source tool that can be used by both optimization practitioners and researchers, while continuing our collaboration with the industry.

2.2 State of the art

2.2.1 Modelling tools

Many different modelling tools exist. Some of them are open source, some commercial. The most commonly used tools for MILP are AMPL, GAMS, JuMP [1] and Pyomo [2]. AMPL and GAMS are two commercial tools that provide their own language and can handle large classes of problems (up to non-linear programming). They interface with various solvers and methods. JuMP and Pyomo are two open-source packages, respectively, embedded in Julia and Pyomo. They can also model up to non-linear programming and interface with various solvers and methods. In their native form, none of these tools support structure encoding or exploits structure in any shape or form. Extensions to these tools exist that support structure encoding such as SAMP[3] for AMPL, EMP[4] for GAMS or Plasmo[5] for JuMP. SAMP extends AMPL to support stochastic programming. It enables the encoding of structures through stages of optimization and interfaces with structure-exploiting methods. EMP is an extension to algebraic modelling languages that reformulates classes of problems such as stochastic programming or disjunctive programming into equivalent problems in other modelling languages such as GAMS by using specific annotations. In its stochastic reformulation, it supports the definition of stages similar to SAMPL. Plasmo enables structure encoding and interfaces with structure-exploiting methods.

2.2.2 Parametric optimization and post-optimal analysis

Below, we propose how to tackle uncertainty in models using various tools, and parametric optimization is one of them. Parametric optimization studies the effect of changing parameters of an optimization problem on its solution. It has a large overlap with techniques called post-optimal analyses. See [6] for a generic and noteworthy review of parametric optimization through the ages. We focus in this proposal on linear parametric optimization, on the left-hand side of the

constraint. The literature generally focuses on very local results around known optimums. Reference [7] is a seminal paper on linear parametric optimization that gives results on the value and derivative of the new optimal solution around non-degenerate optimal bases. Reference [8] is a more recent result which provides (a quite non-tractable) result for degenerate bases.

2.2.3 Bi-level and robust optimization

Bi-level optimization considers optimization problems that are embedded in another optimization problem. A common application of bi-level optimization in energy is energy markets where the upper-level optimization can be the market and the lower-level the operational optimization of each agent in the market such as in [9] or [10].

Robust optimization provides a set of techniques to find solutions sufficiently robust to changes or uncertainties related to the parameters. See [11] for an end-to-end textbook on the subject. There exists quite a few papers and books on the use of robust optimization for energy [12,13,14]. In terms of tools for robust optimization, RSOME [15] is a Python package that enables an easy deployment of a series of robust optimization methods, the definition of uncertainty sets, etc.

2.2.4 Structure exploitation

The performance of decomposition algorithms has already been discussed in literature. In early 2000, Lyu et al. [16] provided an extensive analysis of the speed-ups that can be achieved using parallel Dantzig-Wolfe. In 2010, Gamrath et al. [17] proposed a series of experiments linked to the deployment of Dantzig-Wolfe decomposition given a suitable decomposition and they showed that for structured problems, this method can be very interesting in terms of solving time. Furthermore, Gamrath et al. highlighted the fact that the complexity relies on finding a good partition. In terms of finding a partition, [18] tries to automatically detect hypergraphs in the problem structure in order to generate partitions. Reference [19] discusses the problem of finding a “good” decomposition for Dantzig-Wolfe and proposes machine learning methods to obtain a fully data-driven automatic decomposition framework based on a set of features derived directly from the problem.

2.3 Research project

We divide the project into five work packages based on three main research axes, namely the work on uncertainty, bi-level optimization, and exploiting structure for solving and two auxiliary axes related to the sustainability and integration of the tool.

2.3.1 Work Package 1: Uncertainty

In this subsection we first begin by explaining why uncertainty is an important factor in modelling. We then introduce this WP research subject using a basic mathematical formalisation, that allows practitioners to efficiently tackle the uncertainty related to a partially unknown parameter. We finally explain the general directions the research will take based on this formalization. A significant amount of time has already been spent discussing what can be done and techniques that could be used; this WP will be the first to be tackled and will also define the directions for the other parts of the proposal.

In large-scale energy models, uncertainty is pervasive. It mostly affects parameters of the models: unknown energy or machine prices in the (far) future, unknown weather, unknown energy-related demands and production are the more usual source of uncertainty in this context. Optimal choices provided by models are typically highly dependent on these uncertainties.

These uncertain parameters can simultaneously influence the objective function, the variables’ coefficient in the constraints and the constraints’ biases. A typical task associated with having an uncertain parameter is to assess how the objective value varies with each different choice for the parameter, i.e. to evaluate the following function:

$$\mathrm{f}(\lambda) = \max_{x} c(\lambda)^t x \quad \text{s.t. } A(\lambda)x \geq b(\lambda); x \leq 0$$

Without loss of genericity, we can assume that only the matrix coefficients depend on λ:

$$f(\lambda) = \max_x c^t x \quad \text{s.t. } A(\lambda)x \geq b; x \leq 0$$

The case where λ is a scalar and which influences a single cell of A, is known in the literature as mono-parametric optimization, which was extensively studied in the 1970s. If λ is a vector and influences multiple cells of A, the study of f is known as multi-parametric optimization, and is still a far more open field of research.

In GBOML, a parameter is a well-defined object that can intervene non-linearly into literally anything; the only constraint being that the final model must be (mixed-integer) linear on the variables. The resulting matrix A can thus be non-linear on λ.

A standard way of computing $f(\lambda)$ (when λ is scalar) is to discretize the space into a predefined set of $\lambda_1, \lambda_2, \dots, \lambda_n$ and to simply solve the resulting optimization problems $f(\lambda_1), f(\lambda_2), \dots, f(\lambda_n)$ (so, n times). This is probably inefficient and inaccurate: f may have long periods of stability where recomputing its value is unnecessary, and on the other hand has highly non-linear behaviour necessitating higher resolution sampling. More worryingly: the user cannot know if the resulting approximation of f they obtain is accurate.

This work package is focused on the approximate evaluation of f in a fast and reliable way. We propose to research methods to find the upper and lower bound on f, and to iteratively compute its true value only on points needed to obtain a low estimation error. An idea of the resulting plot for f is provided in Figure 3.

In order to find upper and lower bounds and estimates for f, we propose exploring several possibilities:

- Is it possible to find solutions that are locally robust (i.e., staying solutions) to change to λ? One would provide valid upper bounds (and lower bounds when applied to the dual).
- Is it possible to find "families of solutions" depending on λ, that remain local solutions? As an example, can we find $x = y + \lambda z$ that is a local solution for a range of λ?
- From various optimal conditions, can we efficiently guess the range of λ where an optimal solution remains a valid/optimal solution? This is not trivial since A is non-linearly dependent on λ in general.

We would first focus on the case where A is affine on λ ($A(\lambda) = A^{\text{fixed}} + \lambda A^{\text{var}}$) then on the non-linear, general case.

In terms of additions to the tool, this research will require the addition of robust-optimization-related methods, such as robust constraints (constraints that remain true over a possibly infinite set of scenarios) and chance constraints (constraints that remain true on a large part of a given distribution).

2.3.2 Work Package 2: Bi-level optimization

In many energy problems, we traditionally have two levels of abstraction: the agent/technology level where the actor wants to maximize its economic benefits and the global level which wants to reach a certain goal, typically minimizing the environmental footprint. The hierarchical symbolic framework of GBOML could be used within this context to naturally encode these problems and exploit their structure.

From a purely "tool" perspective, this is already an important endeavour; bi-level linear problems typically require conversion to quadratic problems or MILP to be solved, and the problem is even harder when considering bi-level MILP.

On the research side:

- There is a non-negligible amount of research to be done concerning the language and its API: how to expose these bi-level features to users in an understandable, coherent, and computable way.
- We plan to adapt the techniques developed in the WP1 relating to uncertainty to the bi-level case

2.3.3 Work Package 3: Exploiting structure for solving

The problems in energy sizing and operations which have a physical structure as discussed in Section 1.1. GBOML already interfaces with structure-exploiting methods, namely Dantzig-Wolfe and Benders decomposition. These methods take as input an MILP and a partition of the matrix and return the MILP solution. They have proven to be more efficient in terms of solving time, than classical optimization algorithms on certain problems given “good” partitions.

In the literature, the main approach taken is to find these partitions by working on the constraint matrix. In this work, we want to assess if starting from an already-encoded (often physical) structure can lead to “good” (and hopefully better) partitions.
In the current version of GBOML, we send the plain structure encoded by the user as a partition. The goal of this work package is to find a function that takes as input the plain structure encoded by the user and returns a partition such that the solving time is smaller than the solving time with the plain partition or automatic partition generation based on the structureless matrix of constraints.

In terms of approaches, we will adopt a machine learning or reinforcement learning approach to learn what constitutes a good partition and try to reproduce it given the encoded structure. For this, we will capitalize on the large number of industrial problems already encoded in GBOML to generate datasets of possible partitions, in order to learn from them.

2.3.4 Work Package 4: Results integration into the tool

The various methods developed in WP1/2/3 will be implemented directly in GBOML and made freely accessible. This represents a non-negligeable amount of work, notably to polish and document the new related programming interfaces and methods.

2.3.5 Work Package 5: User support and maintenance

A small (but crucial) share of the research time will be devoted to user support and maintenance of GBOML, to continue to improve its quality, extend its community reach and impact.

2.4 Work plan (to be described for the whole duration of the project)

This project spans over four years:

- The two first years of the project will be devoted to the WP1, i.e., tackling the problem of uncertainty in MILP.
- In year three, we will investigate the implementation of bi-level optimization in GBOML as explained in WP2.
- The fourth and final year will be devoted to exploiting structure in the solving and finding of an optimal partition for Benders decomposition and Dantzig-Wolfe decomposition WP3.

During the whole length of the project, we will work on WP4 and WP5, related to integrating the results into the tool and user support and maintenance. A simple Gantt chart is given in Figure 4.

FIGURES ,TABLES AND REFERENCES FOR SECTION 2

Figure 3 : schema of the tool's output, estimating f.

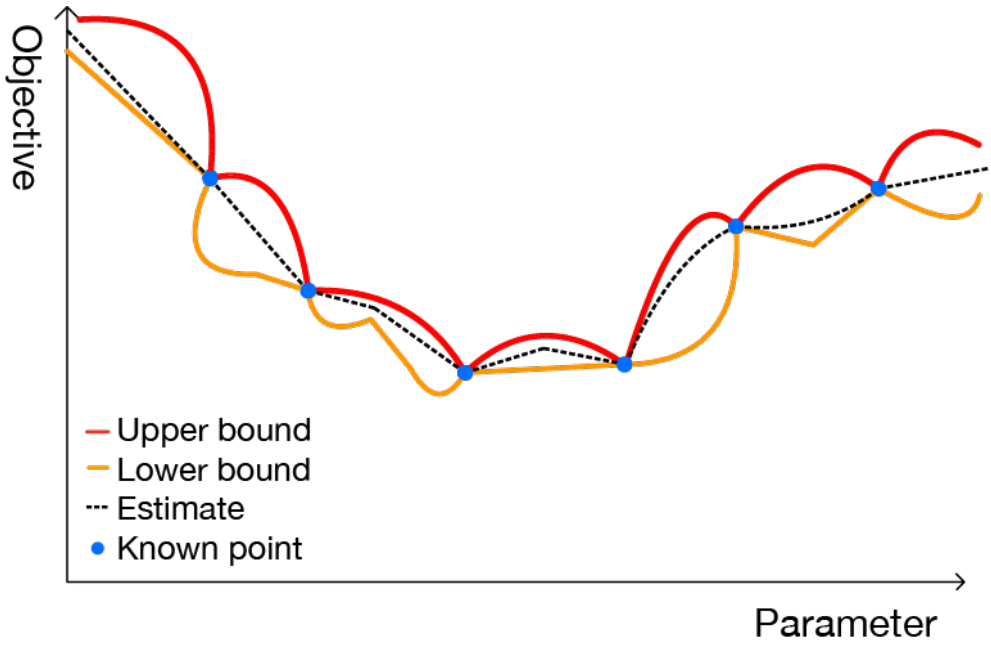

Figure 4 : Gantt chart

	Year 1	Year 2	Year 3	Year 4
Work Package 1: Uncertainty				
Work Package 2: Bi-level optimization				
Work Package 3: Exploiting structure in solving				
Work Package 4: Results integration into the tool				
Work Package 5: User support and maintenance				

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COMMENTS ON CHANGES MADE IN THE RESEARCH PROJECT IN CASE OF RESUBMISSION (OPTIONAL)

In case of former application submitted to the F.R.S.-FNRS via the same funding instrument, please specify the main changes made in your funding application following previous submission, identifying comments from experts that you may have taken into account (max. 1 page).

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3) POTENTIAL INTERDISCIPLINARY APPROACH OF THE RESEARCH PROJECT (OPTIONAL)

If applicable, please identify the interdisciplinary approach of your research project (max. 1 page).

The project will be carried out in the Smart Grids Lab of the Montefiore Research Unit of the ULiège, which focuses on AI (mainly machine learning, reinforcement learning) and energy. While the work in the research project itself has a limited interdisciplinary scope (computer science, language design, mathematical optimization), it has a large interdisciplinary reach within the academic world, and also outside it, through our users at Fluxys and Haulogy and future industrial cooperations.

4) PUBLICATIONS AND ENVIRONMENT OF THE MAIN PROMOTER

Main promoter

Prof. Damien Ernst,
University of Liège
Smart Grids lab

The most significant publications

Please indicate the 10 best publications of your career.

(as measured by citations, source Google Scholar)

- Extremely randomized trees
- Tree-Based Batch Mode Reinforcement Learning
- Reinforcement learning and dynamic programming using function approximators
- Transient stability of power systems: a unified approach to assessment and control
- Active management of low-voltage networks for mitigating overvoltages due to photovoltaic units
- Power systems stability control: reinforcement learning framework
- Contingency filtering techniques for preventive security-constrained optimal power flow
- Reinforcement learning versus model predictive control: a comparison on a power system problem
- Interior-point based algorithms for the solution of optimal power flow problems
- The impact of different COVID-19 containment measures on electricity consumption in Europe

Please indicate the 5 most relevant publications with regard to the project.

- GBOML: Graph-Based Optimization Modeling Language
- Remote Renewable Hubs for Carbon-Neutral Synthetic Fuel Production
- Interior-point based algorithms for the solution of optimal power flow problems
- Jointly Learning Environments and Control Policies with Projected Stochastic Gradient Ascent
- Exploiting the flexibility potential of water distribution networks: A pilot project in Belgium

Please indicate the 5 most important oral communications in your career

- "*Remote renewable energy hubs"*, TotalEnergies, Paris, May 2023
- "*Reinforcement leaning for data-driven optimisation*", EES-UETP Course on Advanced Data Analytics for Energy Systems. Porto, Portugal, September 2018.
- "*The GREDOR project. Redesigning the decision chain for managing distribution networks*", Workshop on "Smart grids energy and big data analytics challenges, opportunities, and solutions: from distribution systems to pan-E
- "*The Global Grid* ", Fête de la Science, RTE, Paris, France, October 2016.
- "*Batch mode reinforcement learning based on the synthesis of artificial trajectories*", Princeton, USA, December 2012.

Scientific environment of the main promoter

Please describe your scientific environment

The Principal Investigator for this research is Prof. Damien Ernst, a well-known figure in the research community owing to his significant contributions in the fields of Artificial Intelligence and Energy Systems. Prof. Ernst has co-authored more than 250 research papers. He has won numerous prizes for his research and, among others, the prestigious Blondel Medal in 2018. Presently, Prof. Ernst heads the Smart Grid group of the ULiège, comprising approximately fifteen researchers, who actively collaborate with various industries and research laboratories.

The Smart Grid group is part of the Montefiore Research Unit which comprises over one hundred active researchers. This research unit specializes in diverse areas such as optimization, energy, neuro-engineering, signal processing, planning under uncertainty, and artificial intelligence, among others. This research project stands to greatly benefit from the extensive network and knowledge-sharing opportunities within this research unit. The GBOML tool, which serves as the focal point of this research, has garnered significant interest from numerous individuals within the unit, further fostering potential collaborative possibilities.

Please describe key elements in your scientific environment which are relevant for the project (e.g.: access to the equipment that are useful for the project, research team, relevant and meaningful collaborations for the project...)

All members of Walloon universities have access to Tier 2 HPC clusters at the CECI, which largely covers our needs for computation capabilities.

Numerous research projects in the laboratory already use GBOML as a day-to-day tool, and so we ensure that we have a constant base of users to report proposals for improvements, bugs and feedbacks. This research project thus integrates nicely into the day-to-day work of the laboratory and will continue to reinforce the collaborations between its internal and external members.

5) PUBLICATIONS OF THE OTHER PROMOTERS
(ONLY FOR MULTI-UNIVERSITY PDR AND PDR WITH PARTNER AGENCIES OF THE WEAVE INITIATIVE)

Other promoter 1 [Table to duplicate as many times as there are promoters]
Full name Institution Laboratory
The most significant publications
Please indicate the 10 best publications of her/his career
Please indicate the 5 most relevant publications with regard to the project.
Please indicate the 5 most important oral communications in her/his career

6) INTERACTIONS AND DISTRIBUTION OF THE TASKS AMONG THE MAIN PROMOTER AND CO-PROMOTERS (ONLY FOR MULTI-UNIVERSITY PDR AND PDR WITH PARTNER AGENCIES OF THE WEAVE INITIATIVE)

Please describe (max. 1 page) the interactions between the different partners of the research programme and be specific on these collaborations, providing details and the distribution of the tasks between the partners.

Non applicable.

7) ADDITIONAL COMMENTS (OPTIONAL)

If you want to communicate elements that have not been mentioned elsewhere in the file, please provide this information below (max. 2 pages).

Please note that in case the presented project provides for the involvement of patients and/or human or animal subjects, it is important that the project includes justifications on the planned sample size (number of subjects included in the study/studies) and how the size is relevant (based on statistical power calculations, for instance). It is also important to explain how the number of patients/subjects expected can be reached. In case the project provides for the involvement of patients and/or subjects, please provide those pieces of information under this section (if not already mentioned elsewhere in the project). Ultimately, this information (or the lack of information) may be taken into account by experts in the frame of the evaluation of your funding application.

Non applicable.